

**AROR UNIVERSITY OF ART, ARCHITECTURE,
DESIGN & HERITAGE, SUKKUR**

Department of Artificial Intelligence

INTERNSHIP REPORT

**IoT Prototype Development for 1:1 Vehicle Simulator Controls
(T85-IIUG Control Layout & SH15 Electrical Operator Panel Digitization Project)**

Submitted by: Tahir Mahmood

Enrolment No: F23-ARI-0050

Internship Organization: Microvision

Duration: 16 June 2026 to 31 July 2026

Submitted to: Faisal Khawaja, Internship Supervisor

Acknowledgment

I would like to thank Microvision for giving me the opportunity to do this internship and gain hands on experience with real world IoT hardware and embedded systems engineering. I am especially thankful to Faisal Khawaja for the guidance, patience, and technical mentorship given throughout the internship period.

I am also grateful to my project teammates, Shah Inayat, Shafique Ahmed, Sania Memon, Ghulam Ali, Tahir Mahmood, and Zuhair Shah, for their collaboration and shared effort on the project planning, panel design, and prototyping work described in this report.

I would also like to thank the faculty of the Department of Artificial Intelligence at Aror University of Art, Architecture, Design and Heritage, Sukkur, especially Raheel Shabbir, for the academic support and for helping arrange this internship as part of my degree requirements.

Executive Summary

This report describes the work done during an internship focused on IoT hardware and embedded systems engineering, carried out as part of a seven member project team. The internship focused on one main technical goal: digitizing physical, electromechanical replicas of vehicle interior control components, including driving levers, toggle switches, rotary dials, and an operator display panel, so that they can drive a PC based simulation environment in real time, styled on a T85-IIUG control layout.

The internship moved through five connected phases. The team first planned a pilot project covering the driving levers (steering bars) and an auxiliary switch panel, then carried out a structured comparison, based on clear criteria, of three candidate microcontroller platforms: the ESP32-S3, the standard ESP32 (WROOM), and the Teensy 4.1. This evaluation led to choosing the Teensy 4.1 with native USB HID for the Sprint 1 prototype, with a documented backup plan using an Ethernet connected ESP32-S3 for panels spread across a network.

On the practical side, I contributed directly to hands on prototyping using the Cirkuit Designer environment. This included writing and successfully compiling ESP32-S3 firmware for non blocking LED sequencing, and testing a two board ESP32-S3 communication circuit using the ESP-NOW protocol. I also researched and put together a verified bill of materials for a companion energy monitoring node. This report explains the objectives, methods, challenges, and design outcomes of this work.

1. Introduction and Company Profile

Modern vehicle simulator training systems rely more and more on digitizing the exact physical controls that a real operator would use, so that the look, feel, and behaviour of every switch, lever, and dial is faithfully copied inside a PC based simulation. This internship was carried out within a team assigned to convert physical, electromechanical replicas of vehicle interior components into a reliable, low latency digital control interface for a simulation environment, following a T85-IIUG control layout.

Company Profile

Microvision (<http://www.microvision.com.pk/>) works in the IT services sector within Pakistan, and specializes in embedded systems, IoT architecture, and high fidelity vehicle simulation and training technologies. Based on the technical scope of work, the internship was placed within a dedicated engineering division focused on the hardware to software digitization pipeline.

2. Internship Objectives and Team Responsibilities

The main objectives of this internship included gaining hands on experience with embedded systems, evaluating microcontroller platforms against technical requirements, and applying safety related design practices such as signal debouncing, electrical isolation, and fail safe interlocks.

Team Structure and Responsibilities

The project was carried out by a seven member engineering team (Muhammad Bilal, Shafique Ahmed, Shah Inayat, Ghulam Ali, Tahir Mahmood, Zuhair Shah, and Sania Memon), reporting to Faisal Khawaja, Internship Supervisor. My specific responsibilities included taking part in technical planning, comparing platforms, developing firmware in Cirqit Designer, and putting together component sourcing reports for the local market.

3. Work Performed and Project Phases

Phase 1: Project Scoping, Driving Bars and Auxiliary Switch Panel

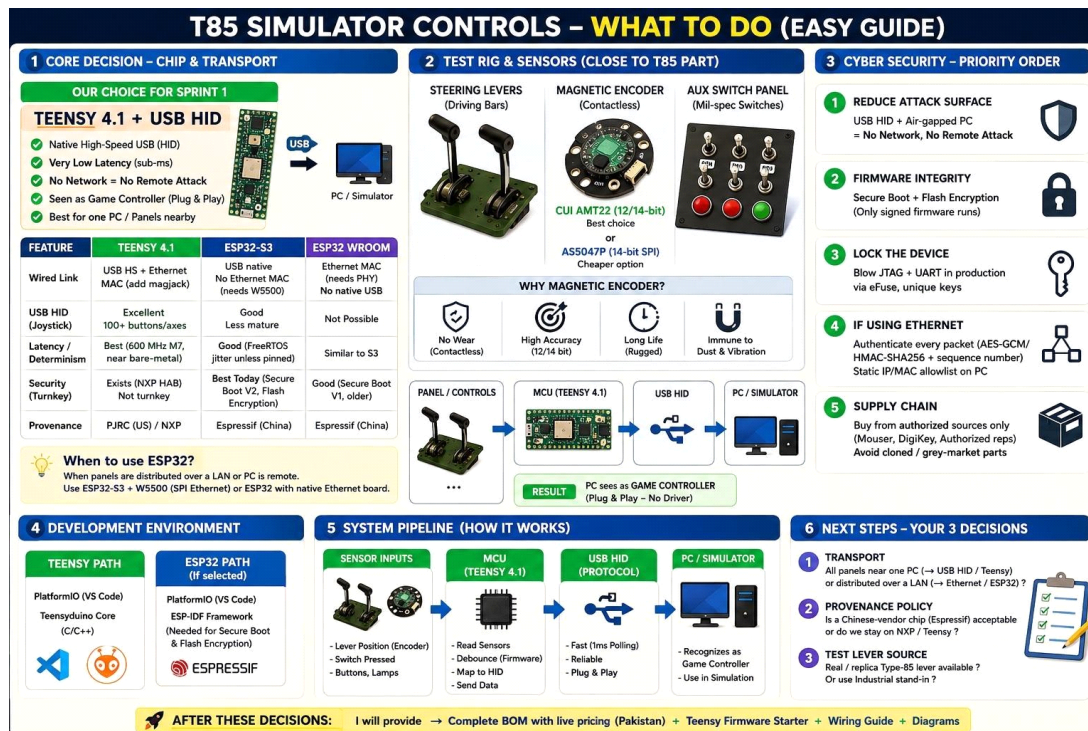
The team started a pilot project covering two key components: the driving levers (steering bars) and an auxiliary switch panel under the T85-IIUG layout. For the driving bars, the team focused on capturing the full range of motion using high resolution absolute rotary encoders or industrial Hall effect potentiometers, without the added complexity of active force feedback. For the switch panel, heavy duty mil spec tactile toggle switches were used, setting a baseline requirement for reliable signal conditioning.

Phase 2: Microcontroller Selection and Justification

Three candidate platforms were carefully evaluated against strict requirements for latency, wired connectivity, and cybersecurity: the ESP32-S3, the standard ESP32 (WROOM), and the Teensy 4.1.

Feature	Teensy 4.1	ESP32-S3	ESP32 (WROOM)
Wired Link	USB High Speed and Native Ethernet MAC	Native USB, no Native Ethernet MAC	Ethernet MAC present, no Native USB
USB HID (Joystick)	Excellent (100+ native axes and buttons)	Good, less mature stack	Not natively possible
Latency and Security	Best (600MHz Cortex M7, bare metal)	Best Hardware Cryptography today	Good security, older architecture

Sprint 1 Strategic Decision: The team chose the Teensy 4.1 with native USB HID for the primary test rig because it completely removes the remote network attack surface and offers plug and play joystick emulation with latency under one millisecond.



4. Detailed Panel Engineering: SH15 Operator Panel

Building on the success of the initial sprint, the team designed a full plan for the complete SH15 Electrical Operator Panel, housed in a mil spec case. The panel works as a smart, standalone device with a dual microcontroller architecture:

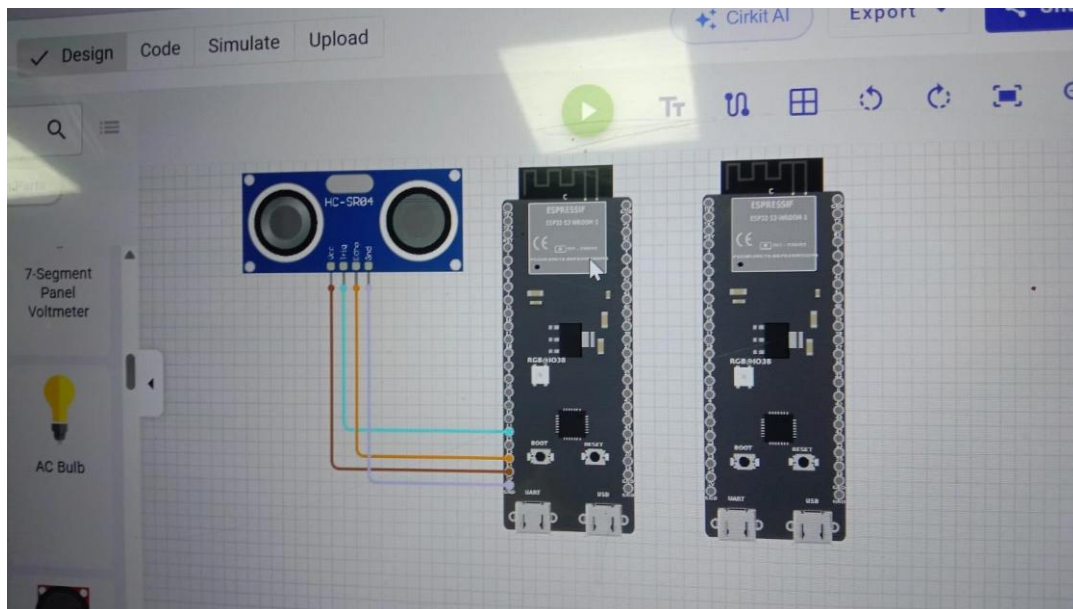
- **Fast, Real Time Controller:** Handles real time switch polling (10 to 20 milliseconds), debouncing, and strict software enforced safety locks (for example, blocking hydraulic commands when Travel Mode is active). Evaluated parts include the STM32G474 and NXP i.MX RT1064.
- **Main Controller:** Drives the diagnostic display screen and handles outbound network serialization. Evaluated parts include the Raspberry Pi Compute Module 4.

Signal Conditioning, Isolation, and Data Flow

To protect host systems, every switch line includes hardware based RC debouncing or dedicated ICs (MC14490) paired with optical isolation (4N35/6N137) before entering the controller. The two internal microcontrollers communicate over a localized Controller Area Network (CAN) bus, sending events instantly along with a required 50 millisecond keep alive heartbeat. A network bridge passes these CAN packets to the main simulator server through secure MQTT.

Core Engineering Parts Inventory

Component Class	Suggested Hardware Type	Engineering Role
Input Switches	Otto / APEM Mil Spec Toggles	Replicate tactile feel of vehicle layout
Isolation / Debounce	4N35 Optocouplers / MC14490 IC	Electrical protection and signal smoothing
Real Time Compute	STM32F407 / Teensy 4.1	High speed polling and safety lock execution
Bus / Network Transceiver	TJA1051 (CAN) + MCP2515 Controller	Multi node localized communication layer



5. Hardware Prototyping and Procurement Sourcing

Practical prototyping was carried out inside the Cirkit Designer workspace, with focus on checking both local logic and board to board messaging pipelines.

ESP32-S3 Non Blocking LED Sequencing Firmware

A state machine based firmware sketch was successfully developed and compiled for the ESP32-S3 Dev Module to sequence status indicators without using the blocking delay() function. This keeps the polling of critical inputs continuous and free of delay. The code compiled successfully with the following footprint:

- **Program Storage Utilization:** 331,055 bytes (25 percent of the 1,310,720 byte maximum limit).
- **Dynamic Memory Allocation:** 20,792 bytes (6 percent of the 327,680 byte maximum limit).

Two Board Communication Rig and Visual Confirmation Pattern

To check reliable communication between devices, two ESP32-S3-WROOM-1 modules were connected with an HC-SR04 ultrasonic distance sensor. The system was programmed using the ESP-NOW protocol. A physical confirmation pattern was designed: the sending board flashes an onboard LED once a packet is delivered successfully, while the receiving board flashes its own LED as soon as a valid packet is received. This matching flashing gave clear visual proof that the connection was stable.

Pakistan Based Energy Monitoring Bill of Materials (BOM)

A second sourcing list was researched and put together for local engineering needs, covering nationwide vendors that offer cash on delivery.

Component Name	Primary Local Vendor	Unit Price (PKR)	Stock Status
ESP32-WROOM-32 Dev Board	Digilog.pk	Rs. 1,090	In Stock
MAX485 TTL to RS485 Module	w11stop.com	Rs. 400	In Stock
Tactile Push Buttons (12mm)	Majju.pk	Rs. 5	In Stock
PZEM-016 AC Energy Meter	Ktechnics.com	Price on Page	In Stock

6. Technical Challenges, Key Outcomes and Recommendations

The main engineering challenge was reducing signal noise on high frequency inputs and preventing possible ground loops between the replica panel and the simulator PC. This was successfully solved by using opto isolation as a standard and planning for hardware based debounce ICs instead of relying only on firmware loops.

Key Technical Recommendations

- **Voltage Standardization:** Formally measure and confirm the real vehicle's native panel supply voltage (24V compared to the 28V military standard) before buying downstream DC to DC converters.
- **CAD Tool Alignment:** Standardize the CAD workflow on either Autodesk Fusion 360 or SolidWorks right away across the four sub teams, to prevent file format mismatches during the assembly of the physical case.
- **Security Hardening:** Apply cryptographic message authentication (AES-GCM or HMAC-SHA256) at the network bridge stage, since the underlying CAN bus works without built in validation layers.